Technical Note: Benchmark Platform Models

Ruixiang Du
ruixiang.du@gmail.com

Abstract

This note documents the platform models implemented in `src/control/models` (namespace conventions, state/control layouts, governing equations, default parameters, and the closed-form oracle each implementation is validated against), with references to the originating literature. The set is chosen deliberately: classical models that are thoroughly understood in the literature yet rich enough to evaluate the estimation, planning, and control algorithms of this repository — each model stresses a property the others do not.

1 Conventions

Every model exposes the discrete `Step` concept used across the stack (`x_(t+1) = "Step"(x_t, u_t, t, Delta t)`, forward Euler unless noted) and, where the dynamics are continuous, the derivative `Deriv` used with the fixed-step classical RK4 propagator (`rk4.hpp`) for high-fidelity simulation. The numerical cores are scalar-templated raw-span functions in `model_core.hpp`, shared verbatim between the CPU implementations and the CUDA rollout programs (see the MPPI note). Units are SI (m, s, rad, N, kg) throughout; frames are stated per model.

2 Wheeled platforms

2.1 Differential drive (kinematic unicycle)

State $\boldsymbol{x} = (x, y, \theta)$ (world position, heading), control $\boldsymbol{u} = (v, \omega)$ (forward speed, yaw rate):

$$\dot{x} = v\cos\theta, \quad \dot{y} = v\sin\theta, \quad \dot{\theta} = \omega \tag{1}$$

The model class of production wheeled MPPI deployments (Nav2); actuator dynamics are absorbed by the box constraints. File: `diff_drive.hpp`.

2.2 Kinematic bicycle (Ackermann)

State $\boldsymbol{x} = (x, y, \theta)$, control $\boldsymbol{u} = (v, \delta)$ (speed, front steering angle), wheelbase L:

$$\dot{x} = v\cos\theta, \quad \dot{y} = v\sin\theta, \quad \dot{\theta} = \left(\frac{v}{L}\right)\tan\delta \tag{2}$$

The standard low-speed car model [1]. Default $L = 0.5$ m (robot scale); the mismatch campaign overrides it to match the dynamic plant. File: `ackermann.hpp`.

2.3 Bicycle with acceleration input

State $\boldsymbol{x} = (x, y, v, \theta)$, control $\boldsymbol{u} = (a, \delta)$:

$$\dot{x} = v\cos\theta, \quad \dot{y} = v\sin\theta, \quad \dot{v} = a, \quad \dot{\theta} = \left(\frac{v}{L}\right)\tan\delta \tag{3}$$

Successor of the retired `model/BicycleKinematics` (wheelbase constant $L = 2.4$ m preserved); the reachability Monte-Carlo propagates it with RK4. **Oracles:** straight-line constant-acceleration closed form; constant-steering circle of radius $L/\tan\delta$. File: `bicycle_accel.hpp`.

2.4 Single-track model with linear tires ("dynamic bicycle")

The standard vehicle lateral-dynamics benchmark (Rajamani [2], ch. 2; the kinematic-vs-dynamic comparison plant of Kong et al. [1]). State $\boldsymbol{x} = \left(X, Y, \psi, v_x, v_y, r\right)$ — world position and heading, **body-frame** longitudinal/lateral velocities, yaw rate; control $\boldsymbol{u} = (a_x, \delta)$. With front/rear axle distances l_f, l_r from the CoM, mass m, yaw inertia I_z, and per-axle cornering stiffnesses C_f, C_r, the linear-tire slip angles and lateral forces are

$$\alpha_f = \frac{v_y + l_f r}{v_x} - \delta, \quad \alpha_r = \frac{v_y - l_r r}{v_x}, \quad F_{yf} = -C_f\alpha_f, \quad F_{yr} = -C_r\alpha_r \tag{4}$$

and the dynamics are

$$\dot{X} = v_x\cos\psi - v_y\sin\psi, \quad \dot{Y} = v_x\sin\psi + v_y\cos\psi, \quad \dot{\psi} = r \tag{5}$$

$$\dot{v}_x = a_x + v_y r, \quad \dot{v}_y = \frac{F_{yf}\cos\delta + F_{yr}}{m} - v_x r, \quad \dot{r} = \frac{l_f F_{yf}\cos\delta - l_r F_{yr}}{I_z} \tag{6}$$

The linear tire model is meaningless near standstill: slip-angle denominators clamp at $v_{x,\min}$ (default 0.5 m/s). Defaults are a mid-size passenger car: $m = 1500$ kg, $I_z = 2500$kg m^2, $l_f = 1.2$ m, $l_r = 1.6$ m, $C_f = C_r = 80000$ N/rad. **Oracle:** the steady-state yaw rate under constant speed and steering,

$$r_{ss} = \frac{v_x}{L + K_{us}v_x^2}\delta, \quad K_{us} = \frac{m}{L}\left(\frac{l_r}{C_f} - \frac{l_f}{C_r}\right), \quad L = l_f + l_r \tag{7}$$

verified to 1% by simulation with v_x held exactly ($a_x = -v_y r$); at low speed the model reproduces the kinematic yaw rate $(v/L)\tan\delta$ to 2%. Role in this repository: the designated **mismatch plant** — planners use the kinematic bicycle, this plant slips (per-seed tire-stiffness variation in the nightly campaign). File: `dynamic_bicycle.hpp`.

3 Underactuated benchmark

3.1 Cart-pole (inverted pendulum on a cart)

The classical underactuated benchmark, in the exact formulation of Barto, Sutton & Anderson [3] (uniform pole, frictionless track and pivot). State $\boldsymbol{x} = \left(x, \dot{x}, \theta, \dot{\theta}\right)$ with θ measured **from upright** ($\theta = 0$ is the unstable equilibrium, $\theta = \pi$ hangs down); control $u = F$ (force on the cart). With cart mass M, pole mass m, pivot-to-CoM distance l (half the pole length):

$$\ddot{\theta} = \frac{g\sin\theta - \cos\theta\cdot\xi}{l\left(\frac{4}{3} - \frac{m\cos^2\theta}{M+m}\right)}, \quad \xi = \frac{F + ml\dot{\theta}^2\sin\theta}{M+m} \tag{8}$$

$$\ddot{x} = \xi - \frac{ml\ddot{\theta}\cos\theta}{M+m} \tag{9}$$

The 4/3 factor carries the uniform pole's moment of inertia ($I_{\text{com}} = m(2l)^2/12$). Defaults are the literature-standard set: $M = 1$ kg, $m = 0.1$ kg, $l = 0.5$ m, force limit ±10 N. **Oracles:** conservation of the mechanical energy

$$E = \frac{1}{2}M\dot{x}^2 + \frac{1}{2}m\Big(\dot{x} + l\dot{\theta}\cos\theta\Big)^2 + \frac{1}{2}m\Big(l\dot{\theta}\sin\theta\Big)^2 + \frac{1}{2}I_{\mathrm{com}}\dot{\theta}^2 + mgl(\cos\theta - 1) \tag{10}$$

under zero force ($|\Delta E| < 10^{-6}$ relative over 10 s of RK4), instability of the upright equilibrium, and bounded oscillation about the hanging one. The integration scenario is the classical controller handover: MPPI swings the pole up globally, a DLQR state-feedback controller (gain synthesized from a numeric linearization at the upright) catches and holds. File: `cartpole.hpp`.

4 Aerial platform

4.1 Quadrotor (rigid body)

The canonical aerial benchmark [4]. State (13): $\boldsymbol{p} \in \mathbb{R}^3$, $\boldsymbol{v} \in \mathbb{R}^3$ (world frame), unit quaternion $\boldsymbol{q}$ (w-x-y-z, body-to-world), $\boldsymbol{\omega} \in \mathbb{R}^3$ in the **body** frame — note the SRB quadruped model (MPPI note) uses world-frame $\boldsymbol{\omega}$; the difference is deliberate and follows each model's literature convention. Control $\boldsymbol{u} = (f, \boldsymbol{\tau})$: total thrust along the body z axis and body-frame moments. With diagonal inertia $\boldsymbol{I}$:

$$\dot{\boldsymbol{p}} = \boldsymbol{v}, \quad \dot{\boldsymbol{v}} = -g\boldsymbol{e}_3 + \left(\frac{f}{m}\right)\boldsymbol{R}(\boldsymbol{q})\boldsymbol{e}_3 \tag{11}$$

$$\dot{\boldsymbol{q}} = \frac{1}{2}\boldsymbol{q} \otimes (0, \boldsymbol{\omega}), \quad \dot{\boldsymbol{\omega}} = \boldsymbol{I}^{-1}(\boldsymbol{\tau} - \boldsymbol{\omega} \times \boldsymbol{I}\boldsymbol{\omega}) \tag{12}$$

`Step` renormalizes the quaternion after the Euler update. Defaults are a lab-scale small quadrotor: $m = 0.5$ kg, $\boldsymbol{I} = \mathrm{diag}(2.32, 2.32, 4.0) \times 10^{-3}$kg m^2. **Oracles:** hover ($f = mg$, level) is an exact equilibrium; free fall matches $z = -gt^2/2$; a pure yaw moment integrates $\omega_z = (\tau_z/I_{zz})t$ exactly (the gyroscopic term vanishes for pure z rotation) with the quaternion norm preserved. File: `quadrotor.hpp`.

5 Legged platform

The single-rigid-body (SRB) quadruped — trunk as one rigid body driven by per-foot ground-reaction forces, massless legs, per-step foot plan and contact schedule — is documented with the MPPI technical note (it is that controller's primary legged plant) and follows the convex-MPC lineage [5]. File: `srb_quadruped.hpp`.

6 Choosing a model

Model	**Stresses**	**Typical use**
double integrator	analytic oracles, LQR baselines	controller unit tests
diff drive / kinematic bicycle	planar planning, box constraints	MPPI wheeled scenarios, tuner
bicycle (accel input)	second-order wheeled, RK4 propagation	reachability Monte-Carlo
dynamic bicycle	**model mismatch**, slip, speed-dependent stability	nightly robustness campaigns

cart-pole	instability, swing-up/catch handover	global+local controller composition
quadrotor	3D attitude, force/moment control	3D estimation + control in the loop
SRB quadruped	contact schedules, GRF control	legged MPPI, GPU rollouts

Bibliography

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